Technische Hochschule Ingolstadt
Fakultät Elektrotechnik und Informatik

REPORT

HiL

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Experiment	*Date of Experiment*	*Date of Submission*
ACC		
CISS		
HIL	*09.04.2026*	*14.04.2026*
RADAR		
SENSOR&ACTUATOR		
STEADY CIRCLE		

1. Preparation and Background

This experiment combined three important topics in vehicle electronics: the CAN protocol, CANoe/CANdb++ as analysis tools, and Hardware-in-the-Loop testing as a safe method for validating ECU behaviour. The task focused on an ESP control unit because it depends strongly on reliable wheel-speed and steering-related signals.

1.1 Controller Area Network (CAN)

Controller Area Network (CAN) is a robust serial communication system used to connect multiple electronic control units in a vehicle. Its main advantage is that many nodes can exchange information over the same bus, which reduces wiring complexity and improves system integration.

In practice, CAN messages are prioritised by their identifiers, so higher-priority messages gain bus access first. This is especially important in safety-related systems such as braking, steering support, and stability control.

1.2 CANoe and CANdb++

CANoe was used in the lab to monitor bus traffic, observe diagnostics, and verify whether injected faults could be seen at communication level. CANdb++ was used to inspect the DBC database, which defines message identifiers, signal locations, factors, offsets, and physical units.

Once a DBC file is loaded, raw hexadecimal data in the trace window can be interpreted as meaningful physical signals. This makes debugging much easier because the engineer can move from byte-level information to system-level understanding.

1.3 Hardware-in-the-Loop Test Environment

The HiL setup consisted of a CarMaker-based vehicle simulation linked to an ESP ECU. This arrangement allowed the ECU to be tested under realistic driving conditions without using a real vehicle. By modifying voltage levels and sensor values, it was possible to analyse how the controller responded to abnormal operating conditions.

2. Experimental Procedure and Results

The experimental work was carried out in a sequence that moved from basic operating checks to deliberate fault injection. The following sections describe what was done, what was observed, and what the results indicate about ECU behaviour.

2.1 Input Voltage Test

The first task examined the effect of supply voltage on the ESP ECU at 12 V, 11 V, 10 V, and 9 V. At 12 V and 11 V, the vehicle largely remained driveable, but the ESP lamp appeared during cornering and disappeared only after a short delay. In both cases, DIAG ERROR 1 still remained visible in the diagnostics window.

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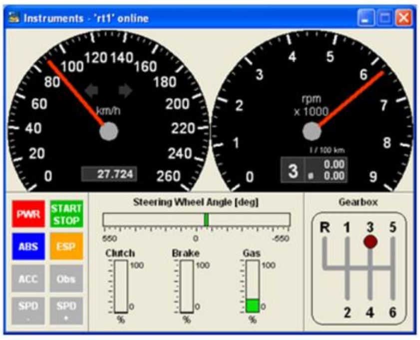
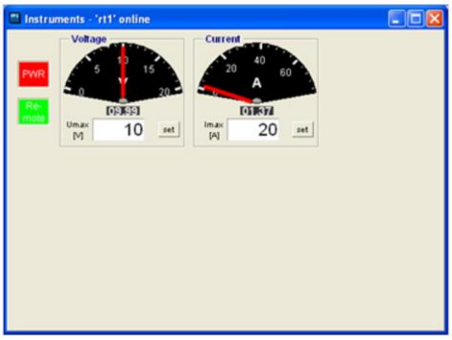
Voltage	Observed vehicle behaviour	Diagnostic status	Interpretation
12 V	ESP lamp appeared mainly during cornering and went off again after a short delay.	DIAG ERROR 1 remained visible.	Near-reference behaviour, but the setup still indicated a stored diagnostic issue.
11 V	Behaviour was similar to the 12 V case. The ESP lamp blinked during dynamic manoeuvres.	DIAG ERROR 1 still remained.	Slight undervoltage did not stop operation, but it did not fully restore a clean diagnostic state.
10 V	Behaviour was similar to 11 V and 12 V, but ABS intervention also appeared during cornering.	DIAG ERROR 1 remained present.	At this point the reduced voltage began to affect system stability more clearly.
9 V	Both ESP and ABS lamps stayed on even while driving straight and while cornering.	DIAG ERROR 1 remained present.	This represents a strongly degraded operating condition caused by severe undervoltage.

When the voltage was reduced to 10 V, the behaviour became more critical because the ABS warning also appeared during cornering. At 9 V, the undervoltage effect was clearly severe: both ABS and ESP stayed on not only in curves but also during straight driving, showing that the ECU had moved into a more protective operating state. This progression confirms that decreasing supply voltage directly affects the reliability of ECU operation and warning behaviour.





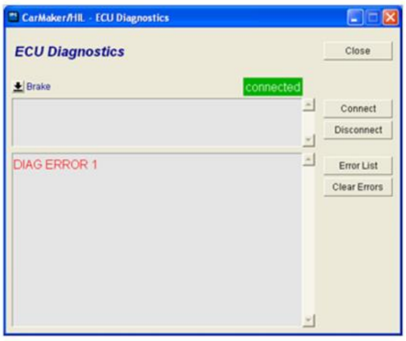


Figure 1. Screenshots recorded during the 10 V test condition: vehicle instruments during cornering, the voltage/current setting window, and the ECU diagnostics view showing DIAG ERROR 1.

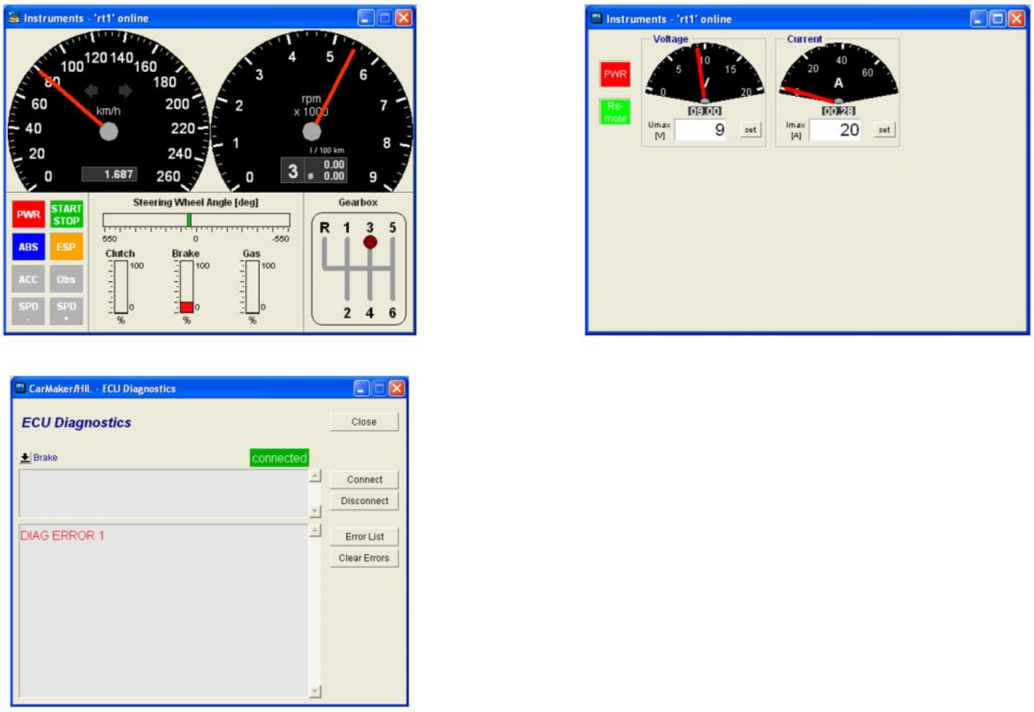


Figure 2. Screenshots recorded during the 9 V test condition. Compared with the 10 V case, the undervoltage state is more critical because both ABS and ESP remain active even during straight driving, while DIAG ERROR 1 continues to be reported.

2.2 Wheel-Speed Signal Fault Injection Using DVA

The next step was to manipulate sensor inputs directly through the Direct Variable Access (DVA) window. First, the front-left wheel-speed signal was forced to zero. During cornering, both ABS and ESP stayed on for a period of time and only cleared again once the vehicle returned to straight-line motion.

This indicates that the ECU recognised the signal as implausible mainly under dynamic conditions, where wheel-speed consistency is more critical.

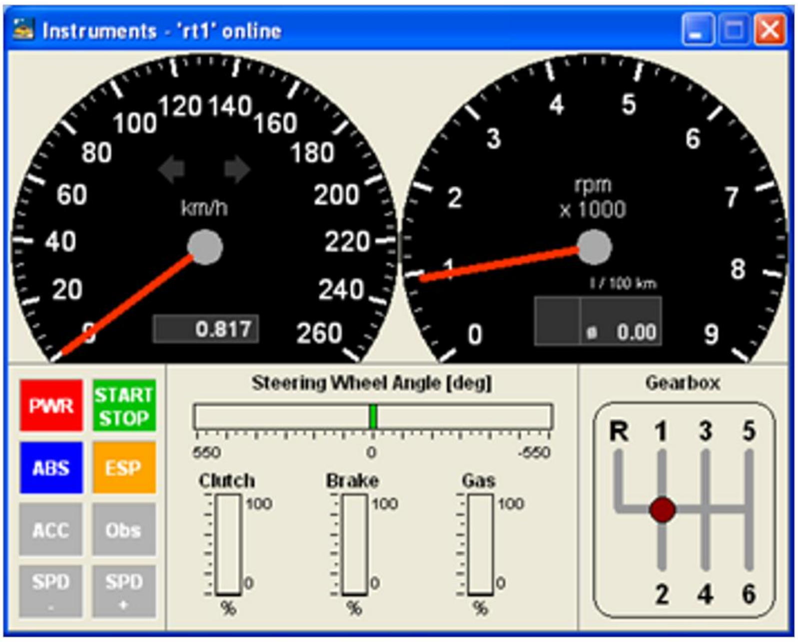


Figure 3. Instrument cluster after setting the front-left wheel-speed signal to zero. The ABS and ESP warning lamps remain active, which confirms that the manipulated input affected the controller response.

After that, three signals were forced to zero simultaneously: IO::WheelSpd_FL, IO::WheelSpd_FR, and IO::WheelSpd_RL. This produced four diagnostic entries in the ECU error list, including one supply-voltage fault and three implausible wheel-speed sensor faults.

Fault code	Meaning
029C 2	Battery supply voltage below the lower limit (undervoltage condition).
0122 8	Rear-left wheel-speed sensor signal implausible.
011D 8	Front-right wheel-speed sensor signal implausible.
011B 8	Front-left wheel-speed sensor signal implausible.

Figure 4. DVA-based fault injection and the resulting ECU diagnostics. The wheel-speed signals were set to zero, and the ECU responded with three implausible-signal faults plus an undervoltage-related entry.

The result shows that the ESP ECU contains plausibility checks for wheel-speed information and can clearly distinguish which sensor channels are affected. This is one of the main benefits of HiL testing: sensor and power faults can be introduced repeatedly and safely, and the resulting ECU response can be evaluated in a controlled environment.

2.3 DBC Inspection and Signal Interpretation in CANdb++

The DBC file was then examined in CANdb++ to understand how the CAN messages were structured. The database provides the relationship between raw message data and physical vehicle signals. Without this information, the trace window would only show hexadecimal bytes; with it, the engineer can identify message names, transmitters, receivers, signal lengths, factors, offsets, and physical units.

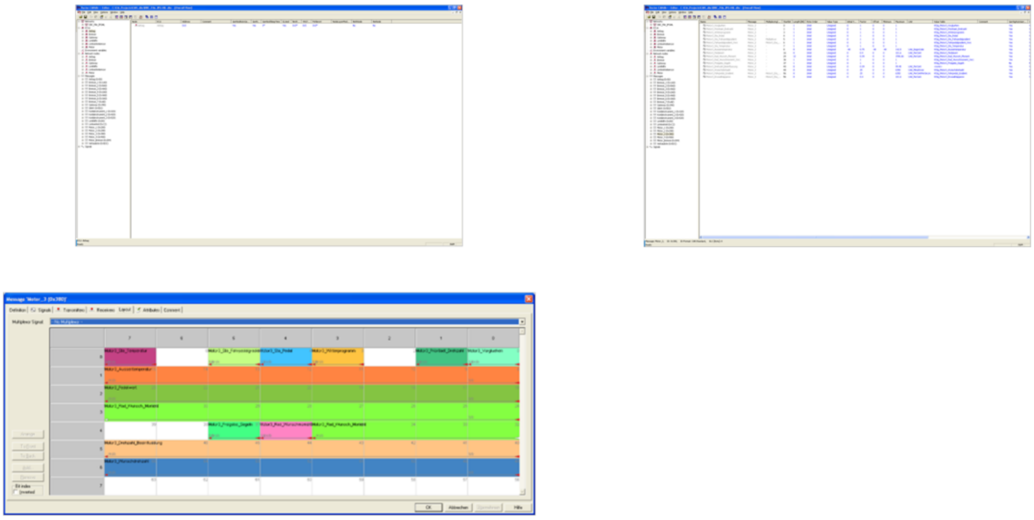


Figure 5. CANdb++ views used during the experiment: database overview with the available ECUs and messages, signal properties after selecting a message, and the bit-level layout of message Motor_3.

The screenshots show that the database includes several ECUs, such as the brake system, gateway, steering-related modules, and motor controller. This structure is essential for network analysis because it tells the user which ECU publishes a signal and how other ECUs should interpret it.

For this reason, loading the DBC file into CANoe is not only convenient but necessary for meaningful analysis. It transforms raw bus traffic into readable engineering information and allows faults to be traced back to the correct message and signal definition.

2.4 Error Observation in CANoe

Finally, the simulated faults were observed directly in CANoe. The trace window showed that the communication problem could be detected at bus level as an ErrorFrame with a Stuff Error entry. This is significant because it demonstrates that CANoe can be used not only for functional signal analysis, but also for identifying protocol-level transmission problems.

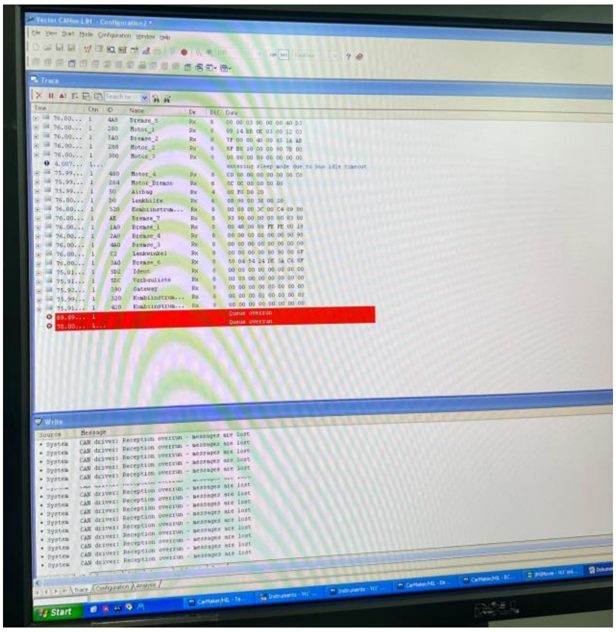


Figure 6. CANoe Trace window showing a simulated ErrorFrame and a reported Stuff Error. The trace confirms that the introduced fault was visible on the CAN bus.

From an engineering perspective, this is useful because it separates two different problem classes: signal plausibility errors at application level and communication faults at protocol level. Both can influence ECU behaviour, but they need different diagnostic approaches.

The additional trace captures provided afterwards showed that the ErrorFrame was reproducible rather than accidental. Across several moments of the run, CANoe again reported Stuff Error entries together with ErrorFrames in the trace window. This repeatability is important because it shows that the communication fault injection could be observed consistently at bus level.

Figure 7. Additional CANoe Trace captures showing repeated ErrorFrame and Stuff Error events at different moments of the simulation. The repeated appearance of these entries indicates that the injected communication fault was consistently visible on the CAN bus and not limited to a single isolated timestamp.

2.5 Statistics, Graphics, and Logged Signal Evaluation

The final screenshot also shows CANoe's statistics and graphics windows, which were used to review the recorded behaviour over time. While the trace window is ideal for frame-by-frame inspection, the statistics view summarises overall bus activity and error-related information, and the graphics window presents selected variables as continuous curves.

This combination is especially useful in Hardware-in-the-Loop testing because it allows the engineer to connect individual communication events with broader signal trends. Instead of relying only on isolated trace entries, it becomes possible to judge whether a signal changes smoothly, whether a disturbance is temporary or persistent, and whether the measured behaviour is repeatable during logged-data evaluation or replay.

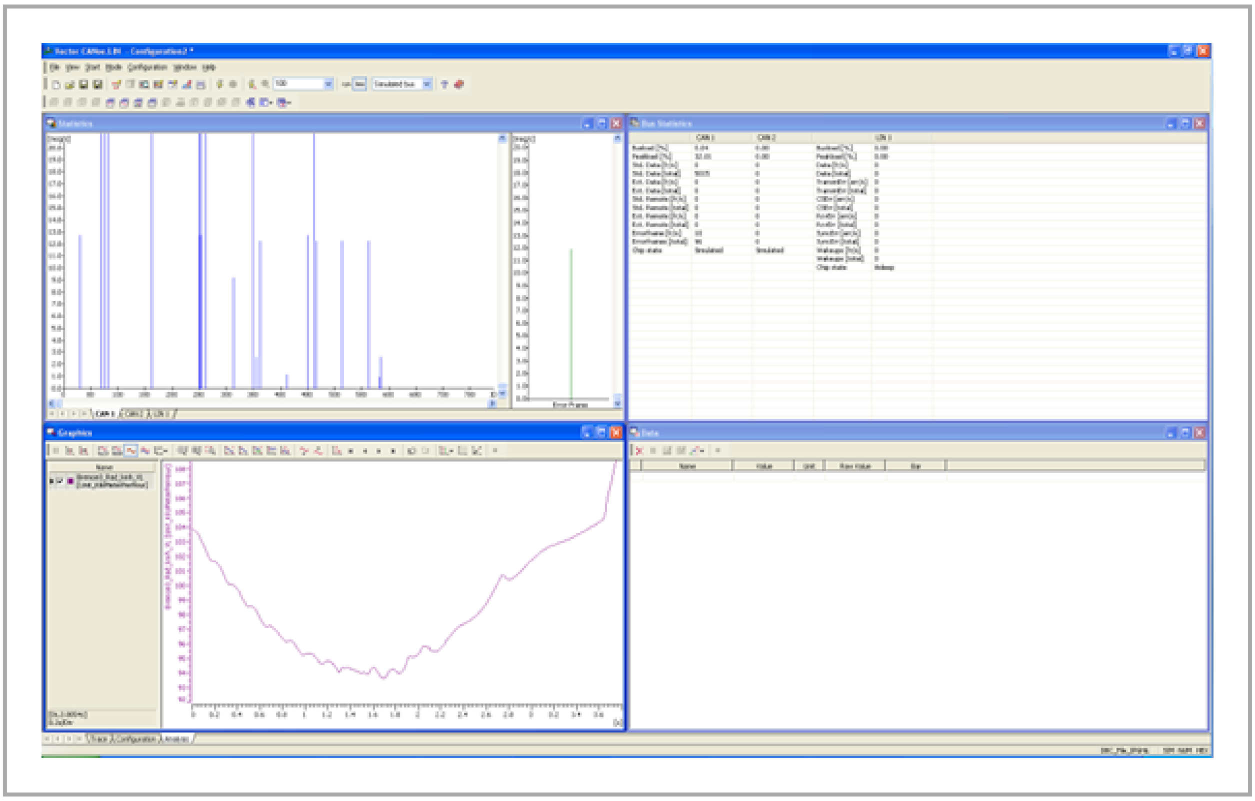


Figure 8. CANoe statistics and graphics view used to evaluate recorded bus activity and selected signal curves. The screenshot complements the trace-based analysis by showing time-based behaviour, overall activity, and the suitability of logged data for later review or replay.

3. Conclusion

This laboratory exercise showed how a Hardware-in-the-Loop setup can be used to evaluate ECU behaviour under realistic but controlled conditions. By changing the supply voltage, injecting sensor faults through DVA, inspecting the DBC structure in CANdb++, and observing the bus in CANoe, the experiment covered both functional and communication-level aspects of vehicle electronics.

The voltage tests demonstrated that reduced supply voltage makes the ESP ECU more conservative, with warning lamps becoming more persistent as the available voltage decreases. The DVA experiments then showed that implausible wheel-speed inputs are detected reliably, and that multiple simultaneous sensor faults lead to clear diagnostic entries for the affected wheels. These results confirm that the ECU contains plausibility checks that are essential for safe operation.

The CANoe trace screenshots, including the additional repeated ErrorFrame captures, also highlighted the difference between application-level signal faults and protocol-level communication errors. Finally, the statistics and graphics windows showed how logged behaviour can be reviewed beyond single trace entries, which is valuable for validation, replay, and repeatability analysis in HiL work.

Overall, the lab provided a strong practical link between CAN theory, automotive diagnostics, and ECU validation. It demonstrated how modern vehicle communication tools are used not only to observe normal system behaviour, but also to inject faults, verify diagnostic reactions, and analyse system robustness in a structured engineering workflow.

End of report